

Algorithm Theoretical Basis Document

Vegetation Index Trend Analysis from Sentinel-2

Mann-Kendall and Sen's slope on seasonal p90 composites

Google Earth Engine implementation
Companion document to `ndvi_trend_mann_kendall_sen.js`

Version 1.1 — NDVI and NDMI support
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1. Introduction

1.1 Purpose and scope

This document describes the theoretical basis, input data, processing steps, outputs, and known limitations of a Google Earth Engine (GEE) tool for the automated analysis of vegetation trends. The tool computes per-pixel trend statistics in one or more spectral vegetation indices — currently the Normalised Difference Vegetation Index (NDVI) and the Normalised Difference Moisture Index (NDMI) — over the active vegetation season of a user-defined Area of Interest (AOI), using the full Sentinel-2 archive from 2017 to the present. The trend is summarised through the non-parametric Mann-Kendall test and the Sen's slope estimator, applied to a series of annual seasonal composites.

The tool is designed for multi-partner deployment. Each project partner can apply it to their own AOI and define their own active vegetation season, while the analytical pipeline, the input data products, and the quality-control procedures remain identical across partners. This ensures that trend products generated by different partners are methodologically comparable.

1.2 Intended audience

This document is intended for project partners who will run the script on their own AOIs, and for reviewers and stakeholders who need to understand and assess the algorithm. It assumes familiarity with remote sensing, Sentinel-2 data, and basic statistical concepts. Operational instructions on how to run the script in the Code Editor are out of scope; they are kept in the script header.

1.3 Document structure

Section 2 summarises the algorithm and lists inputs and outputs. Section 3 describes the input data products and their relevant characteristics. Section 4 details the theoretical basis of each processing step. Section 5 describes the outputs in detail. Section 6 covers quality control, uncertainty, and known limitations. Section 7 specifies the parameters that must be harmonised across partners and those that can be locally adapted. Section 8 lists the references.

2. Algorithm overview

The algorithm transforms a dense, irregular, and partially clouded time series of Sentinel-2 reflectance images into a small number of high-quality annual composites of one or more vegetation indices, on which a per-pixel trend test is performed. The conceptual core is the reduction from many noisy intra-seasonal observations to a single robust value per pixel per year, followed by a non-parametric trend analysis on the resulting annual series.

2.1 Inputs

Input	Description	Notes
AOI	Polygon or MultiPolygon, as FeatureCollection asset, Feature, or drawn geometry	Multiple polygons are dissolved for the pixel-wise analysis; per-feature statistics are produced separately
Indices	One or more of: NDVI, NDMI	Computed independently; results are kept in separate output bands
Active vegetation season	Two integers: start and end day-of-year (DOY)	Set by the partner based on local phenology
Year range	Two integers: first and last year	Sentinel-2 SR starts on 2017-03-28; 2018 is recommended as first year for a homogeneous series
QC parameters	Cloud Score+ threshold, minimum scenes per season, minimum valid years, significance level	Defaults provided; harmonisation across partners is recommended

2.2 Outputs

Output	Type	Content
Trend result	Multi-band image, ONE PER INDEX, exported to Drive (GeoTIFF) or Asset	Sen's slope, Sen's slope significant-only, Kendall's tau, Z, p-value, S, valid_years
Annual composites	Single-band images, one per year per index, exported to Drive (GeoTIFF) or Asset	Seasonal 90th-percentile of each requested index
Per-feature time series	Interactive chart in the Code Editor	Mean p90 NDVI per year and per AOI feature
Area summary	Console printout	AOI area covered by data, by significant positive trend, by significant negative trend

2.3 Processing workflow

The processing chain is organised in the following steps:

- Load and normalise the AOI; display it on the map for visual verification.
- Build the Sentinel-2 collection over the AOI and within the seasonal DOY window, link the matching Cloud Score+ scenes.
- Mask each scene using Cloud Score+ and compute NDVI.
- For each year, reduce the seasonal NDVI stack to a 90th-percentile composite and count the number of valid observations per pixel.
- Compute the per-pixel count of valid annual composites (the per-pixel sample size).

- Build all pairwise comparisons between annual composites; from these, derive the Mann-Kendall S statistic and the set of pairwise slopes.
- Compute the variance of S, the standardised Z statistic, the two-sided p-value, Kendall's tau, and Sen's slope.
- Mask pixels with too few valid years; export the multi-band result (one file per index) and the annual composites; print area statistics.

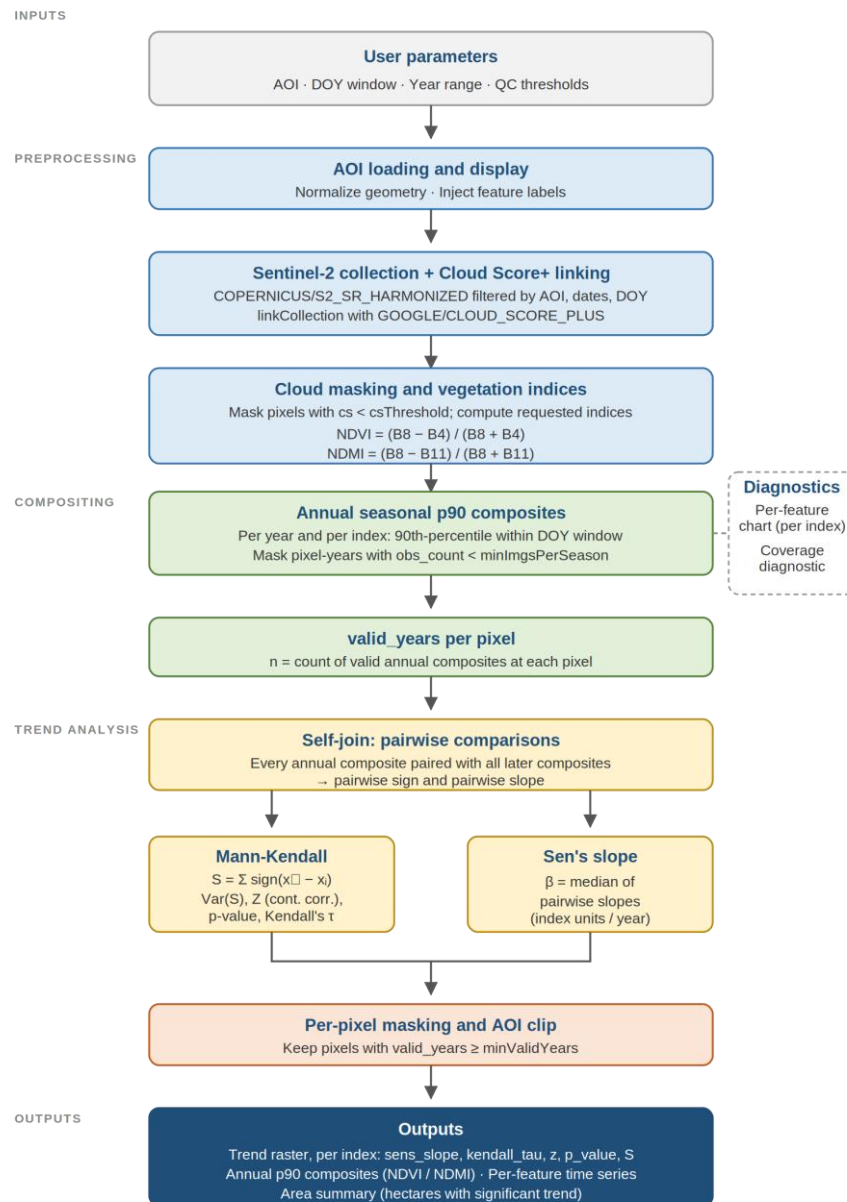


Figure 1. Processing workflow of the NDVI trend analysis algorithm. Solid arrows indicate the main data flow; the dashed callout represents diagnostic outputs produced alongside the analytical chain.

3. Input data

3.1 Sentinel-2 Surface Reflectance Harmonised

The reflectance source is the Sentinel-2 MultiSpectral Instrument (MSI) Surface Reflectance product, distributed in the GEE catalogue as `COPERNICUS/S2_SR_HARMONIZED`. This collection provides Bottom-of-Atmosphere (BOA) reflectance at 10 m for the visible and near-infrared bands used here (B4 and B8), and is corrected for atmospheric effects through the Sen2Cor processor. The harmonised version compensates for the +1000 digital number offset introduced in processing baseline 04.00 on 25 January 2022; failing to use the harmonised collection introduces a spurious step in any inter-annual analysis crossing that date.

The Sentinel-2 archive used here begins on 28 March 2017 (S2A) with S2B operational from mid-2017 onward. Revisit frequency over a given AOI is 2-5 days depending on latitude and tile overlap. The S2C satellite, operational from late 2025, increases revisit frequency further but is fully integrated into the same collection.

3.2 Cloud Score+

Cloud and shadow masking is performed using Cloud Score+ (`GOOGLE/CLOUD_SCORE_PLUS/V1/S2_HARMONIZED`), a machine-learning-based cloud and shadow quality product released by Google Earth Engine in 2023. It assigns to every Sentinel-2 pixel a continuous score in the range 0 to 1, where 1 indicates a fully clear observation and lower values reflect increasing likelihood of cloud, shadow, or other surface artefacts. The product provides two bands: `cs` (per-pixel score) and `cs_cdf` (the cumulative-distribution-function transformation of `cs` over time, slightly smoother).

Cloud Score+ is now widely considered the most reliable cloud and shadow product for Sentinel-2 and substantially outperforms the legacy QA60 band and the Scene Classification Layer (SCL), particularly for thin clouds, cirrus, and shadows in topographically complex terrain. It is linked to the Sentinel-2 image collection by the `system:index` property using the `linkCollection()` method.

3.3 Area of Interest

The AOI must be a polygon or multipolygon. Points and lines are not accepted, since the analysis is intrinsically pixel-wise and pixel-wise statistics over an area require a two-dimensional support. The AOI may be provided as a `FeatureCollection` asset (the typical case of an uploaded shapefile), as a single `Feature`, or as a polygon drawn directly in the Code Editor; all three are normalised internally.

If the input `FeatureCollection` contains several features, they are dissolved into a single geometry for the pixel-wise analysis. The per-feature NDVI time-series chart, however, treats each feature individually. There is no upper limit on AOI extent in principle, but computational performance and the risk of rate-limited tile rendering scale with AOI area; for areas above a few thousand hectares, exporting the result (to Drive or to a GEE asset) and visualising the exported file is recommended over on-the-fly visualisation.

3.4 Active vegetation season

The active vegetation season is specified by two day-of-year (DOY) integers, inclusive, representing the start and end of the period during which the partner expects the vegetation in the AOI to be in active growth. Using DOY rather than month-day allows windows that do not coincide with whole calendar months and ensures consistent treatment across years. The current implementation assumes the season does not cross 1 January, which holds for all northern-hemisphere AOIs targeted by the project.

The seasonal window is the most important phenological choice made by each partner. It should capture the active growing phase of the dominant vegetation in the AOI, including its peak greenness, while avoiding shoulder periods dominated by bare soil, senescent vegetation, or residual snow. Where partners have access to local phenology data (e.g. MODIS-derived land-surface phenology, ground observations, or expert knowledge), these should inform the choice of doyStart and doyEnd.

4. Algorithm theoretical basis

4.1 Cloud and shadow masking

Each Sentinel-2 scene in the collection is masked using its linked Cloud Score+ band, retaining only pixels with cs greater than or equal to a user-defined threshold (default 0.60). All other pixels are removed before any further computation, including NDVI calculation. This ensures that NDVI is never computed on cloudy, shadowed, or thin-cloud-contaminated pixels.

The default threshold of 0.60 is the value suggested as a reasonable starting point by the GEE documentation and corresponds to retaining moderately clear observations. Lower values (0.40-0.50) admit more pixels but include more thin-cloud contamination, which biases NDVI low and tends to depress the percentile composite. Higher values (0.70-0.80) produce cleaner observations but increase the risk that some pixel-years end up with too few valid observations for a reliable composite. Sensitivity to this threshold should be assessed at least once on a representative AOI before fixing it as the project-wide default.

4.2 Vegetation indices

The algorithm computes one or more spectral vegetation indices from the cloud-masked Sentinel-2 reflectance. Two indices are currently supported: the Normalised Difference Vegetation Index (NDVI) and the Normalised Difference Moisture Index (NDMI). Each requested index is computed independently and propagated through the rest of the pipeline as a separate band, allowing partners to analyse a single index or compare results across indices on the same AOI.

4.2.1 NDVI — Normalised Difference Vegetation Index

NDVI is computed from the near-infrared (B8, ~842 nm) and red (B4, ~665 nm) bands using the standard formulation:

$$NDVI = (B8 - B4) / (B8 + B4)$$

NDVI is sensitive to the amount and density of chlorophyll-containing vegetation cover. It is a robust and widely adopted proxy for vegetation greenness and biomass. The output is a floating-point band ranging from -1 to +1, with dense green vegetation in active growth typically falling in the 0.6-0.9 range under Sentinel-2 BOA reflectance. NDVI saturates over very dense canopies (closed broadleaf forests, vigorous crops at peak growth), where small changes in canopy structure produce only small NDVI variations. As a consequence, NDVI tends to respond to vegetation stress only when it has already produced visible canopy changes, such as defoliation or chlorosis.

4.2.2 NDMI — Normalised Difference Moisture Index

NDMI is computed from the near-infrared (B8, 10 m, ~842 nm) and the first short-wave infrared (B11, 20 m, ~1610 nm) bands:

$$NDMI = (B8 - B11) / (B8 + B11)$$

NDMI is sensitive to the equivalent water thickness of vegetation canopies, that is, to the amount of liquid water in plant tissues. The SWIR-1 band is strongly absorbed by liquid water, so a decrease in vegetation water content lowers B11 reflectance and increases NDMI; conversely, water-stressed canopies show lower NDMI. Because vegetation water content responds to soil moisture deficit before chlorophyll degradation becomes visible, NDMI tends to detect drought and water stress earlier than NDVI. The same formula is referred to in the literature as the Land Surface Water Index (LSWI) in agronomy and wetland studies, and as the Normalised Difference Water Index (NDWI) of Gao in forestry; the present document uses NDMI throughout for consistency.

Since B11 has a native resolution of 20 m and B8 a native resolution of 10 m, NDMI is effectively a 20 m product; GEE resamples B11 to 10 m on the fly when the two bands are combined, but the underlying information content remains at 20 m. This is acceptable for the trend analysis performed here, which already integrates over time and is dominated by intra-annual rather than fine-scale spatial variability.

4.3 Seasonal 90th-percentile composite

For each year in the analysis window, the masked index scenes falling within the active vegetation season are aggregated to a single per-pixel value. The aggregation statistic is the 90th percentile, computed with `ee.Reducer.percentile([90])`. The same compositing rule is applied independently to each requested index.

The choice of the 90th percentile is a compromise between three considerations. The seasonal maximum is the standard greenness indicator in phenological studies but is highly sensitive to residual cloud, snow, and high-value outliers; the seasonal mean is smoother but is depressed by any cloud-mask leakage and dominated by lower-value shoulder observations; the 90th percentile retains the upper-tail signal of peak season while being robust to a small number of contaminated observations. It does not depend on the length of the seasonal window (unlike integrated indices), which makes it directly comparable across AOIs with seasons of different duration.

An important caveat applies when the number of valid observations within a season is small. The 90th-percentile estimator on five observations is essentially the maximum; on ten observations, it is the second-highest value. As the observation count varies between pixel-years, the effective statistic represented by p90 varies as well. This is partially controlled by the minimum-observations filter described in the next section.

4.4 Per-pixel quality control

Quality control is applied at two levels. First, for each pixel-year, the number of valid (cloud-masked) observations within the seasonal window is computed; pixel-years with fewer than minImgsPerSeason observations (default 5) are masked out. This prevents extremely sparse pixel-years from contributing unstable p90 estimates.

Second, after the annual composites are built, the count of valid annual composites is computed per pixel; pixels with fewer than minValidYears valid annual composites (default 6) are masked out from the trend statistics. This ensures that the Mann-Kendall and Sen's slope statistics are only computed on time series long enough to provide a meaningful trend signal. Pixels at the very edge of an AOI, where Sentinel-2 tile coverage may be incomplete, or in persistently cloudy locations, are removed by this filter.

4.5 Mann-Kendall trend test

The Mann-Kendall (MK) test (Mann 1945; Kendall 1975) is a non-parametric test for the presence of a monotonic trend in a time series. It makes no assumption on the distribution of the data, requires only that observations be independent, and is robust to outliers. It is a standard tool in hydrological and ecological time-series analysis and is the default trend test in many environmental monitoring frameworks.

Given an annual time series $x_1, x_2, \dots, x_n$ for a pixel, the MK statistic S is the sum, over all pairs of years $i < j$, of the sign of the difference $(x_j - x_i)$:

$$S = \text{sum over all } i < j \text{ of } \text{sign}(x_j - x_i)$$

where $\text{sign}(d)$ is +1 if $d > 0$, -1 if $d < 0$, and 0 if $d = 0$. Positive S indicates an upward trend; negative S a downward trend. Under the null hypothesis of no trend, and for n greater than approximately 10, S is approximately normally distributed with mean zero and variance:

$$\text{Var}(S) = n(n-1)(2n+5)/18$$

in the absence of tied values. Ties produce a correction term subtracted from the variance, but exact ties in floating-point NDVI percentile composites are essentially impossible, and the no-ties formula is used here. The standardised statistic Z is obtained with a continuity correction:

$$Z = (S - \text{sign}(S)) / \text{sqrt}(\text{Var}(S)) \quad \text{for } S \neq 0, \quad Z = 0 \text{ for } S = 0$$

The implementation computes S by performing a self-join of the annual composites: each annual composite is paired with all later annual composites through `ee.Join.saveAll` on the `system:time_start` property. For each pair, the per-pixel sign of the difference is computed; the S statistic is then the per-pixel sum of these signs across all pairs.

4.6 Sen's slope estimator

The Sen's slope (Sen 1968), also known as the Theil-Sen estimator, is the non-parametric counterpart of the ordinary-least-squares slope. It is defined as the median of all pairwise slopes between observations:

$$beta = median\ over\ all\ i < j\ of\ (x_j - x_i) / (t_j - t_i)$$

where $t_j - t_i$ is the time difference between observations j and i (in this case, the year difference). The estimator is robust: up to roughly 29% of the observations may be outliers without unduly affecting the slope. It is therefore particularly well-suited to environmental time series with occasional anomalous years.

The same set of pairwise comparisons used to compute the MK S statistic is reused to compute the pairwise slopes. The Sen's slope is then obtained as the per-pixel median of the pairwise slopes through `ee.Reducer.median()`, expressed in units of NDVI per year.

4.7 Statistical significance

Under the null hypothesis of no trend, Z is approximately standard normal for sufficiently large n . The two-sided p-value is therefore:

$$p = 2 (1 - \Phi(|Z|))$$

where Φ is the standard normal cumulative distribution function. GEE does not provide a native standard-normal CDF, so it is implemented through the Zelen and Severo (1964) polynomial approximation, which is accurate to about $7.5e-8$ in absolute terms over the relevant range and is sufficient for the significance levels of interest (0.01 to 0.10).

The asymptotic-normal approximation underlying this p-value is well-established for n greater than approximately 10. For shorter series (the typical case here, with n between 7 and 9), it is biased toward apparent significance, and exact tables of the S distribution should be preferred when statistical rigour is required. The asymptotic p-value is retained here for practical reasons (per-pixel computation server-side), but partners are encouraged to interpret p-values close to the significance threshold with caution.

4.8 Kendall's tau

As a complement to S and the p-value, Kendall's rank correlation coefficient is computed as:

$$tau = S / (n (n - 1) / 2)$$

Tau ranges from -1 (perfectly decreasing series) to +1 (perfectly increasing series) and measures the strength of the monotonic association independently of the significance test. It is robust to the small- n significance bias discussed in the previous subsection and should be inspected alongside the p-value when interpreting results.

5. Outputs

5.1 Trend result raster

The Mann-Kendall test and Sen's slope estimator are applied independently to the annual series of each requested vegetation index. NDVI and NDMI describe different canopy properties (chlorophyll-based greenness and equivalent water thickness, respectively) and behave differently in their statistical distribution; treating them as separate analyses is methodologically required and is reflected in the output: one raster file is produced per index, not a single combined file. Each per-index raster contains the following bands at 10 m resolution, masked to the AOI and to pixels with at least minValidYears valid annual composites:

Band	Type	Description
sens_slope	Float	Sen's slope, in index units per year (all pixels with enough data)
sens_slope_significant	Float	Sen's slope masked to pixels where the trend is statistically significant ($p_value \leq pSignificance$); non-significant pixels are no-data and appear transparent in the GeoTIFF
kendall_tau	Float	Kendall's tau, dimensionless, in $[-1, +1]$
z	Float	Standardised MK statistic, with continuity correction
p_value	Float	Two-sided asymptotic p-value, in $[0, 1]$
S	Float	Mann-Kendall S statistic
valid_years	Float	Per-pixel number of valid annual composites

The two output bands `sens_slope` and `sens_slope_significant` are deliberately redundant. The first preserves the full information content of the trend analysis, allowing partners to re-thresholds significance at any level in post-processing. The second is a ready-to-use product for map display and area statistics at the project-wide significance level ($pSignificance = 0.05$ by default), where unmasked pixel values directly indicate the magnitude and sign of the significant trend.

5.2 Annual composite rasters

One single-band raster per analysis year per requested index is exported, either to Google Drive as a GeoTIFF or to a GEE asset (configurable via the `exportTo` parameter). Each raster contains the seasonal 90th-percentile of the index at 10 m resolution, masked to pixels with at least minImgsPerSeason valid observations in that year. These outputs allow partners to inspect individual annual composites, perform additional analyses outside the trend framework (for example, comparison with reference vegetation indices or with ancillary land-cover layers), and support visual interpretation of the trend result.

5.3 Per-feature NDVI time series

If the AOI FeatureCollection contains multiple features, an interactive line chart is printed in the Code Editor console, showing one line per feature: the spatial mean of the seasonal p90 NDVI for that feature, plotted against year. This chart functions as a sanity check on the trend analysis: a monotonic visual trend in the chart should correspond to a coherent spatial pattern in the trend raster, and conversely, the absence of any visual trend in the chart should temper expectations of finding significant per-pixel trends.

5.4 Coverage diagnostic

Before computing the trend statistics, the script prints the minimum, maximum, mean, and median of the per-pixel valid_years count across the AOI, alongside the theoretical maximum ($\text{endYear} - \text{startYear} + 1$). If the mean is substantially below the theoretical maximum, the quality-control filters are masking out a non-trivial fraction of pixel-years; this is the first diagnostic to consult when the trend result appears unexpectedly sparse.

5.5 Area summary

The final printout in the console reports three quantities, in hectares: the area of the AOI on which the trend statistics could be computed (sufficient valid years), the area with significant positive trend at the chosen significance level, and the area with significant negative trend. These are the headline summary statistics for the AOI.

6. Quality control, uncertainty, and known limitations

6.1 Sample size and statistical power

With Sentinel-2 SR available from 2017 onwards, the maximum length of the annual series is approximately 8-9 values (depending on the choice of startYear). At this sample size, the Mann-Kendall test has limited statistical power: only relatively strong and consistent trends produce significant p-values. A non-significant result should be interpreted as the absence of evidence for a trend rather than evidence of absence. Conversely, a significant result on such a short series may reflect a single anomalous year as much as a structural change; sensitivity analyses are recommended.

6.2 Asymptotic versus exact p-values

The p-value implementation is the asymptotic-normal approximation. For n less than 10, exact tables of the Mann-Kendall S distribution (see Hipel and McLeod 1994) provide more accurate inference. Partners with stringent statistical requirements may wish to recompute p-values offline using the exported S and valid_years bands together with an exact-distribution lookup.

6.3 Influence of anomalous years

Single anomalous years (typically drought years, such as 2022 in much of Europe) can influence Sen's slope substantially when n is small. Sen's slope is robust by construction, but with only 7-9 points an anomalous value still affects roughly a quarter of the pairwise slopes. Partners are advised to inspect their annual composite series, identify any years with strong

continental-scale climate anomalies, and where relevant produce a complementary trend analysis with the anomalous year excluded as a robustness check.

6.4 Stability of the 90th-percentile composite

The 90th percentile is a stable statistic when the number of valid observations is sufficient. With fewer than five valid observations in a season the p90 effectively reduces to the maximum and becomes more sensitive to high-NDVI noise (residual cloud edges, sun-glint, snow patches). The `minImgsPerSeason` filter prevents the most extreme cases, but partners working in cloudy regions or with very narrow seasonal windows should monitor the `obs_count` band exported in the annual composites to verify that the effective sample size per pixel-year is adequate.

6.5 Sensitivity to Cloud Score+ threshold

The trend result has some sensitivity to the Cloud Score+ threshold. At a fixed AOI, lowering the threshold tends to admit more thin-cloud-contaminated observations, which depress p90 in cloudy years and may produce spurious negative trends; raising the threshold reduces this risk but may exclude entire pixel-years in cloudy regions. A one-time sensitivity test on a representative AOI, comparing the trend result at thresholds of 0.50, 0.60, and 0.70, is recommended before fixing the project-wide value.

6.6 Multi-feature AOIs

When multiple features are provided in the AOI, the pixel-wise analysis is performed on the dissolved geometry, and the trend raster does not retain feature identity. The per-feature time-series chart preserves identity, but per-feature aggregated trend statistics (e.g. mean Sen's slope per polygon, or the fraction of significant area per polygon) are not currently produced. They can be obtained as a post-processing step by `reduceRegions` over the exported trend asset and the AOI `FeatureCollection`, if needed by individual partners.

6.7 Spatial autocorrelation

The area statistics reported as headline summaries treat each pixel as an independent unit. Sentinel-2 NDVI is in practice spatially autocorrelated at the scale of vegetation patches, so the effective number of independent samples is smaller than the pixel count. This does not affect the per-pixel trend statistics but should be borne in mind if partners derive higher-level inferences (for example, hypothesis testing on the fraction of AOI under significant trend).

6.8 Joint interpretation of NDVI and NDMI

When both indices are computed, the most informative reading of the results comes from their joint pattern rather than from each index in isolation. NDVI and NDMI respond to different physical canopy properties — chlorophyll-based greenness and equivalent water thickness — and their relationship to drought, land-use change, and phenological shift differs. Four broad patterns are useful to keep in mind.

A coherent decline in both NDVI and NDMI typically indicates a sustained loss of vegetation: vegetation removal, severe and persistent drought-driven dieback, or conversion to sparser cover types. A coherent increase in both is consistent with vegetation establishment, regrowth, or intensification of agricultural production.

A decline in NDMI without a corresponding decline in NDVI is the signature most relevant to an early-warning framework. The canopy is losing internal water content but has not yet shed leaves or lost chlorophyll. This pattern often precedes drought-induced visible damage by weeks to months and may reveal subclinical stress that NDVI alone would miss, particularly in dense canopies where NDVI is close to saturation.

A decline in NDVI without a corresponding decline in NDMI suggests a change in canopy structure or composition that is not water-driven: deciduous phenology shift, harvest, defoliation by pests, or land-use conversion to a non-vegetated but moist surface. Conversely, an increase in NDMI without a corresponding NDVI signal can mark a transition toward wetter surface conditions, including the recovery of soil moisture or the expansion of wet vegetation around groundwater discharge areas — a pattern of direct interest in spring-monitoring contexts.

Operationally, partners are encouraged to overlay the Sen's slope maps for the two indices, identify pixels where the two slopes diverge in sign or in magnitude, and treat those areas as priority candidates for further investigation. The per-feature time-series charts also help: NDVI and NDMI lines that move in parallel confirm a coherent biophysical change, while divergent lines flag patterns of stress or transition that justify closer inspection.

7. Standardisation across partners

7.1 Parameters to harmonise

To ensure comparability of trend products generated by different partners, the following parameters should be fixed at the same value across the consortium:

Parameter	Recommended default	Rationale
csThreshold	0.60	Cloud Score+ threshold; to be confirmed by sensitivity analysis on representative AOIs
minImgsPerSeason	5	Minimum valid scenes per pixel-year; balances data retention and composite stability
minValidYears	6	Minimum valid annual composites per pixel; ensures meaningful trend test
pSignificance	0.05	Significance level for area statistics and display
Compositing statistic	90th percentile	Robust to residual cloud and noise; independent of seasonal window length

7.2 Parameters to set locally

The following parameters must be set by each partner based on local conditions:

- **AOI:** the partner's own area, as polygon or multipolygon asset.
- **indices:** one or more of NDVI and NDMI. NDVI alone is sufficient for greenness trends; NDMI alone is appropriate when the focus is on canopy water content; running both is recommended whenever early-warning of water stress is of interest, since their joint pattern is more informative than either index alone (see Section 6.8).
- **doyStart, doyEnd:** the active vegetation season for the AOI, in DOY, based on local phenology.
- **startYear:** 2018 is recommended for a homogeneous series; 2017 only if the local season starts after the Sentinel-2 archive becomes available for the AOI.
- **endYear:** the most recent year for which the season has fully ended.
- **aoiLabelProperty:** the property (attribute column) of the AOI FeatureCollection to use as label for each feature in the time-series chart, for example 'name', 'plot_id', 'sito'. If left null, the system:index is used as fallback.
- **exportTo:** the export destination, one of 'drive', 'asset', or 'both'. The default is 'drive', which writes GeoTIFFs to a folder in the partner's Google Drive; 'asset' stores them as Earth Engine assets for further server-side processing; 'both' produces both copies.

- **driveFolder:** the name of the destination folder in Google Drive (used when exportTo includes 'drive'). The folder is created automatically if it does not exist. Each partner should choose a folder name that identifies the run and the AOI.
- **assetFolder:** the GEE asset folder path (used when exportTo includes 'asset'), e.g. 'users/<username>/SentinelSprings'. The folder must already exist in the Assets tab.
- **exportCRS:** the output projection for the exported files. Leave null (default) to use the native Sentinel-2 UTM projection of the AOI, which preserves square 10 m pixels and is the strongly recommended setting. Override only if a specific CRS is required by downstream tools, e.g. 'EPSG:3035' (LAEA Europe) for pan-European products. Avoid 'EPSG:4326', which yields rectangular pixels at mid-latitudes (e.g. about 10 m by 7 m at 45 deg N).
- **chartYMin, chartYMax:** y-axis range for the time-series chart. Default null on both lets Google Charts auto-scale to the data range, which is appropriate for the narrow ranges typical of dense vegetation (e.g. NDVI 0.78-0.91). Set explicit numbers, e.g. 0 and 1, to force the full theoretical index range; you can set only one and leave the other auto-scaled.

7.3 Recommended workflow

Before running the full analysis on a new AOI:

- Run sections 2 (AOI loading) and 6 (coverage diagnostic) only, to confirm that the QC filters are not over-masking the AOI.
- If mean valid_years is substantially below the theoretical maximum, consider relaxing minImgsPerSeason or the Cloud Score+ threshold for diagnostic purposes, and identify the source of low coverage (cloud regime, AOI at tile edge, seasonal window too narrow).
- Inspect the per-feature time-series chart (one chart per requested index) for a visual sense of each index before producing the trend raster.
- Export the trend result and annual composites to Drive (default) or to a GEE asset; visualise the exported files, not the on-the-fly computation, for any further work or for sharing screen captures.

8. References

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